
PROMPT INTERVENTIONS FOR MITIGATING DIALECT PERFORMANCE GAPS IN LARGE LANGUAGE MODELS

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ABSTRACT

Though modern large language models (LLMs) exhibit strong performance on reasoning tasks expressed in Standard American English (SAE), significant performance gaps emerge when these tasks are expressed in underrepresented English dialects under standard prompting. This hands-on project investigates whether simple inference-time prompt interventions—dialect-aware instruction framing, dialect normalization (paraphrasing dialectal input into SAE), and few-shot dialect exposure—can reduce these gaps without additional training. Using the EnDive benchmark (Gupta et al., 2025) as a testbed, we evaluate GPT-4o-mini, GPT-4o, and Gemini 2.5 Pro on four reasoning tasks (WSC, MultiRC, LogicBenchMCQ, and FOLIO) expressed in African American Vernacular English (AAVE) and Indian English (IndE). Our results show that dialect performance gaps are task- and dialect-dependent, most pronounced for WSC IndE (up to 30 pp for Gemini 2.5 Pro, 12.9 pp for GPT-4o), with logic tasks showing consistent but smaller gaps (1-6 pp). Of the three interventions, few-shot prompting is the most consistently beneficial, improving dialect accuracy and achieving full or near-full gap closure in several conditions. Dialect normalization yields gains on reading comprehension tasks but severely degrades accuracy on tasks sensitive to precise wording (e.g., -18.5 pp on WSC IndE for GPT-4o). Dialect-aware instruction framing is highly model-dependent: Gemini 2.5 Pro benefits substantially on WSC AAVE (100% gap closure), while GPT-4o-mini degrades on several tasks. Taken together, these results suggest that LLM dialect robustness requires solutions beyond lightweight prompting. Our code is publicly available on GitHub.

1 INTRODUCTION

As large language models become increasingly deployed in real-world NLP applications, ensuring their robustness across the diverse linguistic backgrounds of users is a central fairness concern. English alone encompasses hundreds of regional, social, and cultural dialects (Blodgett et al., 2020), yet LLMs are predominantly trained on Standard American English (SAE) text. Recent work has documented significant performance drops when reasoning tasks are expressed in underrepresented English dialects rather than SAE, raising concerns about equitable access to NLP capabilities (Gupta et al., 2025; Lin et al., 2025; Hofmann et al., 2024).

The EnDive benchmark (Gupta et al., 2025) systematically evaluated five LLMs across 12 NLU and reasoning tasks translated into five underrepresented English dialects and found consistent accuracy gaps even under chain-of-thought prompting. ReDial (Lin et al., 2025) and related work confirm that these gaps are persistent and that simply asking models to rephrase dialectal input into SAE does not fully close them. However, no study has systematically compared a suite of prompt-level interventions—instruction framing, normalization, and few-shot exposure—specifically designed to mitigate dialect reasoning gaps. If such interventions are effective, they would represent a low-cost, training-free mitigation strategy. If they are not, this points to deeper architectural limitations requiring retraining or fine-tuning.

In this project, we conduct a systematic evaluation of three prompt interventions on four reasoning tasks across two dialects and three models. Our main findings are: (1) dialect performance gaps are task- and dialect-dependent, largest for WSC IndE (up to 30 pp for Gemini 2.5 Pro, 12.9 pp for GPT-4o), with logic tasks showing consistent but smaller gaps (1–6 pp); (2) few-shot prompting is the most effective and robust intervention, achieving full gap closure in multiple conditions (e.g., WSC AAVE across all models, LogicBenchMCQ IndE for GPT-4o and Gemini 2.5 Pro); (3) dialect normalization is high-variance and task-sensitive, providing large gains in some conditions (e.g., WSC AAVE for Gemini 2.5 Pro: +9.0 pp), while severely harming others (e.g., WSC IndE for GPT-4o: −18.5 pp); and (4) dialect-aware instruction framing is strongly model-dependent, with Gemini 2.5 Pro responding most positively (e.g., WSC AAVE: 100% gap closure), while GPT-4o-mini shows degradation on several tasks (e.g., LogicBenchMCQ IndE: −7.7 pp). These results collectively suggest that closing dialect gaps in LLMs requires more than inference-time prompt engineering.

2 RELATED WORK

Dialect variation in NLP. English is not monolithic: it encompasses hundreds of regional, social, and cultural varieties that differ systematically in phonology, morphology, syntax, and lexicon. NLP systems trained predominantly on Standard American English (SAE) text have long been known to underperform on non-standard varieties (Blodgett et al., 2020). Recent high-profile work by Hofmann et al. (2024) demonstrated that LLMs exhibit covert racism triggered by dialect features alone: models assign speakers of African American English to less-prestigious jobs and are more likely to convict and sentence them to death, even when race is never explicitly mentioned. Critically, human preference alignment (RLHF) was found to exacerbate the discrepancy between overt and covert stereotypes, masking prejudice at the surface level while leaving deeper biases intact. Pan et al. (2025) further showed that specific grammatical features characteristic of non-standard dialects—such as existential “it,” zero copula, and “y’all”—explain the majority of observed performance degradation, suggesting that dialect bias is concentrated in particular linguistic structures rather than uniformly distributed across dialect vocabulary and syntax. This finding directly motivates our task selection: WSC and formal logic tasks are precisely the settings where such structural features are most likely to interact with task-critical linguistic cues.

Dialect benchmarks. Several benchmarks have been developed to systematically measure dialect-related performance disparities. Multi-VALUE (Ziems et al., 2023) introduced a rule-based morphosyntactic transformation system covering 50 English dialect features, enabling stress tests on tasks such as question answering and semantic parsing. DialectBench (Faisal et al., 2024) scaled this effort to 281 language varieties across 10 NLP tasks, providing the first large-scale cross-dialectal evaluation. EnDive (Gupta et al., 2025) specifically targets LLM reasoning, translating 12 tasks from SAE into five underrepresented English dialects (AAVE, Indian English, Jamaican English, Chicano English, and Colloquial Singaporean English) using few-shot prompting validated by native speakers, with average human-evaluation scores $\geq 6.02/7$. We use EnDive as our primary testbed because its controlled translation methodology and breadth of reasoning tasks make it well-suited for evaluating prompt interventions under a fixed experimental protocol. ReDial (Lin et al., 2025) complements EnDive by focusing on AAVE with human-authored parallel queries across seven reasoning benchmarks, providing an independent validation of the gaps EnDive identifies.

Prompt engineering for bias mitigation. A parallel line of work explores whether prompt-level techniques can mitigate bias in LLMs without retraining. Approaches include self-debiasing through controlled decoding (Schick et al., 2021), cultural-identity prompting, affective priming, and structured multi-step pipelines that detect and rewrite biased outputs. However, these techniques have primarily targeted demographic biases (e.g., gender, race) in generation tasks rather than dialect-induced performance gaps in reasoning (Kumar et al., 2025). Notably, Lin et al. (2025) found that simply asking models to rephrase AAVE queries into Standard English does not close the performance gap—a finding that directly motivates our work. Where ReDial tests a single normalization strategy in isolation, we systematically compare three qualitatively distinct interventions (instruction framing, normalization, and few-shot exposure) across multiple tasks and models, allowing us to identify not just whether prompting helps on average, but *when* and *for whom* each strategy suc-

ceeds or fails. This granularity is essential for practical deployment: a practitioner needs to know which intervention to apply to which task, not just whether interventions work in aggregate.

Positioning of our work. While prior work has extensively documented that dialect performance gaps exist (Gupta et al., 2025; Lin et al., 2025; Faisal et al., 2024; Ziems et al., 2023) and that prompt engineering can reduce certain biases, no study has systematically evaluated multiple prompt intervention strategies (instruction framing, normalization, and few-shot exposure) specifically designed to close dialect reasoning gaps, nor measured the conditions under which each intervention harms rather than helps. We use EnDive as a controlled testbed to quantify when such strategies help, fail, or introduce undesirable semantic drift, and we situate our findings in the broader context of what closing dialect gaps would require if lightweight prompting proves insufficient.

3 PROBLEM SETUP

We study the following question:

Can simple, low-cost prompt interventions substantially reduce the performance gap that LLMs exhibit on dialectal English inputs, without retraining or fine-tuning?

Inputs. Each input is a reasoning task instance from the EnDive benchmark (Gupta et al., 2025), provided in both SAE and one of two target dialects, African American Vernacular English (AAVE) and Indian English (IndE). We selected AAVE and IndE because they represent two linguistically and socioculturally distinct dialect families with large global speaker populations (approximately 30 million and 129 million speakers, respectively), and because both have received attention in prior fairness work (Lin et al., 2025; Hofmann et al., 2024).

We consider four prompting conditions for each dialectal input: (1) a **baseline** condition using the original EnDive prompt, (2) a **dialect-aware instruction** that explicitly tells the model which dialect to expect, (3) a **dialect-normalization** pipeline that paraphrases dialectal inputs into SAE before feeding them to the model, and (4) a **few-shot** condition that prepends examples of dialectal inputs paired with correct answers. The full prompt templates are described in Section 4.

Outputs. For each task instance, the model produces a predicted answer. We compare this prediction against the ground truth label to compute accuracy.

Motivation. If dialect performance gaps are “fragile” (that is, sensitive to simple prompt modifications) then lightweight prompt engineering could serve as a practical, scalable mitigation strategy deployable at inference time. If the gaps are “robust” to such interventions, this suggests deeper architectural or training-data limitations that require more fundamental solutions.

Goals. In this evaluation, our goals are (1) to quantify whether prompt interventions reduce the SAE–dialect accuracy gap, and (2) to determine whether certain dialects or task types are more prompt-sensitive than others.

Model selection. We evaluate three LLMs: GPT-4o-mini, GPT-4o, and Gemini 2.5 Pro. These models were selected to span two model families (OpenAI and Google) and two capability tiers (GPT-4o-mini as a cost-efficient baseline, GPT-4o and Gemini 2.5 Pro as stronger models), allowing us to assess whether intervention effects generalize across providers and capability levels.

4 METHOD

Each run of our evaluation pipeline executes one trial of a given task-dialect-model-condition combination. First, task instances and their ground-truth labels are loaded from the corresponding dataset in the EnDive benchmark. Next, for each task instance, the dialectal task input is sent to the model via an API call under one of four prompting conditions:

Baseline. The original prompt with no additional framing. This replicates the standard evaluation from Gupta et al. (2025).

Intervention 1: Dialect-aware instruction. We append to the beginning of the instruction: “*You are given a question written in [dialect under test].*” This tests whether explicitly cueing the model to expect dialectal variation improves robustness.

Intervention 2: Dialect normalization. We use the target model to paraphrase each dialectal input into SAE, before sending the normalized input to the target model in a separate call. This tests whether removing dialect features recovers SAE-level performance.

Intervention 3: Few-shot dialect exposure. We prepend a small number ($N=3$) of worked examples showing dialectal input paired with the correct answers. This tests whether in-context examples of dialect-answer mappings help the model generalize.

Each response is parsed to extract the predicted answer, scored against the ground truth, and logged in JSON format. The overall task accuracy is reported at the end of each trial.

Contributions. Our work is novel in that it systematically compares multiple prompt-level intervention strategies on a controlled dialectal benchmark, measuring both their effectiveness and their potential harms across tasks, dialects, and model capability levels. While prior work has documented dialect performance gaps (Gupta et al., 2025; Lin et al., 2025; Faisal et al., 2024) and explored prompt engineering for general bias mitigation (Schick et al., 2021), no study has evaluated prompt interventions specifically designed to close dialect reasoning gaps. Our code is publicly available on GitHub.

5 EXPERIMENTAL SETUP

Tasks. We evaluate on four reasoning tasks from the EnDive benchmark across two categories. For **Language Understanding**, we use the Winograd Schema Challenge (WSC) (Levesque et al., 2012; Wang et al., 2019) and Multi-Sentence Reading Comprehension (MultiRC) (Khashabi et al., 2018). For **Logic**, we use LogicBench MCQ (Xu et al., 2024) and FOLIO (Han et al., 2024). WSC tests commonsense-based pronoun disambiguation; MultiRC tests multi sentence evidence integration; LogicBenchMCQ tests formal logical inference; and FOLIO tests first-order logic reasoning over premises and hypotheses.

The Winograd Schema Challenge (WSC) dataset contains instances of pronoun disambiguation tasks. Given a short passage, the task is to determine whether a pronoun used in the passage refers to the given proper noun. The pronoun resolution challenge is designed to be simple for humans but difficult for models that rely purely on statistical patterns like word associations, testing a model’s commonsense understanding abilities (Levesque et al., 2012; Wang et al., 2019).

The Multi-Sentence Reading Comprehension (MultiRC) dataset includes multi-sentence passages associated with questions and multiple potential answer choices, of which multiple may be correct. Each instance requires the model to determine whether a given answer choice is a correct answer to the given question. The questions were constructed to require information gathered from multiple sentences in the passage in order to be answered correctly, testing a model’s multi-sentence reasoning abilities (Khashabi et al., 2018).

The LogicBench dataset contains symbolic reasoning problems spanning propositional, first-order, and non-monotonic logics, designed to test a model’s ability to carry out formal logical inference rather than pattern matching (Xu et al., 2024). The LogicBench MCQ subset instantiates each problem as a multiple-choice question, where the model must select the logically valid conclusion from several candidates.

The FOLIO dataset includes instances of first-order logic reasoning problems. Each task requires the model to determine the correctness of a given conclusion based on a set of premises, testing the model’s logical reasoning capabilities (Han et al., 2024).

Dialects. We focus on AAVE and Indian English (IndE), chosen for their linguistic distinctiveness, speaker population size, and the availability of prior evaluation baselines.

AAVE is a systematic variety of American English primarily spoken by many Black Americans, characterized by its own phonology, grammar, and vocabulary that differ in regular ways from SAE (Blodgett et al., 2020; Hoffman, 2024).

IndE is a set of English varieties spoken in India and the Indian diaspora, influenced by contact with South Asian languages and widely used in government and education (Ziems et al., 2023; Faisal et al., 2024).

Evaluation metrics. We selected the following evaluation metrics for direct measurement and comparison of model performance across the SAE and dialectal conditions.

1. **Task accuracy:** the primary performance metric, computed as the percentage of correct predictions.
2. **Accuracy gap vs. SAE:** the difference $\text{Acc}_{\text{SAE}} - \text{Acc}_{\text{Base}}$, measuring the magnitude of the pre-intervention performance disparity between the SAE and baseline dialect conditions (i.e., under the direct prompting condition). Positive values indicate worse performance on dialectal input, negative values indicate worse performance on SAE input.
3. **Performance change (Δ , pp):** the change in dialect accuracy under an intervention relative to the baseline dialect condition, defined as

$$\Delta = \text{Acc}_{\text{Intervention}} - \text{Acc}_{\text{Base}}.$$

This metric directly reflects the absolute gain or loss in accuracy for dialectal inputs. Positive values indicate improved performance under the intervention, while negative values indicate degradation.

4. **Gap closure (%):** the relative reduction in the accuracy gap after applying an intervention, defined as

$$\frac{(\text{Acc}_{\text{SAE}} - \text{Acc}_{\text{Base}}) - (\text{Acc}_{\text{SAE}} - \text{Acc}_{\text{Intervention}})}{\text{Acc}_{\text{SAE}} - \text{Acc}_{\text{Base}}} \times 100.$$

This metric captures each intervention’s effectiveness reducing the original disparity between SAE and dialectal input. A value of 100% indicates complete closure of the gap (parity with SAE performance), while 0% indicates no improvement. Positive values indicate reduced disparity under the intervention, while negative values indicate increased disparity. For interpretability, we cap values at $\pm 100\%$.

Evaluation protocol. For each task–dialect–model combination, we compute the task accuracy metric separately under all four prompting conditions and compare the resulting SAE–dialect accuracy gaps. This allows us to assess whether any intervention meaningfully reduces (or amplifies) the performance gap relative to the baseline prompt. For the detailed evaluation pipeline, see Section 4.

6 RESULTS

Table 1 reports baseline accuracy (%) and SAE–dialect gap (pp) for all three models. Results are from our own zero-shot evaluation.

Several patterns emerge from the baseline results:

- **WSC shows the largest gaps across all models.** The gap is most pronounced for Gemini 2.5 Pro on IndE (30.0 pp) and GPT-4o on IndE (12.9 pp). Notably, Gemini 2.5 Pro shows a substantially smaller gap on AAVE (7.0 pp) than on IndE, suggesting the model is more sensitive to Indian English dialect features in pronoun disambiguation tasks. GPT-4o-mini shows negligible WSC gaps because its overall WSC accuracy ($\sim 53\%$) is near chance, leaving little room for dialect-related degradation.
- **The WSC IndE gap scales with model capability.** For IndE, gap magnitude increases across model tiers: GPT-4o-mini (2.7 pp), GPT-4o (12.9 pp), and Gemini 2.5 Pro (30.0 pp). Stronger models appear to rely more heavily on syntactic disambiguation cues that IndE

Table 1: Baseline accuracy (%) and SAE–dialect gap (pp) under standard prompting. Positive gap values indicate worse performance on dialect. GPT-4o-mini and GPT-4o were evaluated on the full set of tasks from the EnDive benchmark (Gupta et al., 2025). Gemini 2.5 Pro was evaluated on a random sample ($N = 200$) of tasks from EnDive due to computational time and cost limits.

Task	Dialect	GPT-4o-mini			GPT-4o			Gemini 2.5 Pro		
		SAE	Dialect	Gap	SAE	Dialect	Gap	SAE	Dialect	Gap
WSC	AAVE	53.4	53.1	0.3	59.7	52.8	6.9	87.0	80.0	7.0
WSC	IndE	52.3	49.5	2.7	58.9	46.0	12.9	89.0	59.0	30.0
MultiRC	AAVE	81.8	79.2	2.6	86.0	85.6	0.4	94.0	93.0	1.0
MultiRC	IndE	80.9	81.6	−0.8	83.5	83.8	−0.3	91.0	89.0	2.0
LogicBenchMCQ	AAVE	76.0	72.5	3.5	79.3	76.0	3.3	90.0	84.0	6.0
LogicBenchMCQ	IndE	74.8	72.2	2.6	75.2	74.3	0.9	84.0	82.0	2.0
FOLIO	AAVE	56.8	56.0	0.8	66.8	65.2	1.6	80.0	78.0	2.0
FOLIO	IndE	56.4	57.0	−0.7	73.8	70.1	3.7	85.0	81.0	4.0

may disrupt, producing larger gaps. This scaling is less pronounced for AAVE, where GPT-4o (6.9 pp) and Gemini 2.5 Pro (7.0 pp) show nearly identical gaps.

- **LogicBenchMCQ shows consistent moderate-to-large gaps.** Gaps range from 0.9–3.5 pp for GPT-4o-mini and GPT-4o, but rise to 6.0 pp for Gemini 2.5 Pro on AAVE, suggesting that stronger models are not immune to dialect-induced degradation on formal logical inference tasks.
- **MultiRC and FOLIO IndE show flat or negative gaps.** GPT-4o-mini and GPT-4o perform comparably or slightly better on IndE dialect than SAE for MultiRC, indicating that not all task-dialect combinations require mitigation and that interventions applied in these settings risk actively harming performance.

6.1 INTERVENTION RESULTS

Tables 2, 3, and 4 show dialect accuracy (%) under the baseline (direct) and three intervention (dialect-aware, paraphrase, and few-shot) prompting conditions for GPT-4o-mini, GPT-4o, and Gemini 2.5 Pro respectively, with baseline SAE accuracy included for reference. These tables report both absolute performance and the corresponding change relative to the baseline dialect condition (Δ , pp).

Table 5 reports gap closure (%) for each intervention, measuring the extent to which the intervention reduces the performance disparity between SAE and dialectal inputs.

Table 2: Dialect accuracy (%) for GPT-4o-mini under all four prompting conditions. Baseline SAE shown for reference. Δ denotes change from baseline dialect accuracy (pp). Bold denotes improvements of 3.0 pp or more above the baseline dialect accuracy.

Task	Dialect	SAE	Base.	D-Aware (Δ)	Para. (Δ)	F-Shot (Δ)
WSC	AAVE	53.4	53.1	51.5 (−1.6)	54.0 (+0.9)	60.0 (+6.9)
WSC	IndE	52.3	49.5	49.5 (0.0)	48.5 (−1.0)	55.6 (+6.1)
MultiRC	AAVE	81.8	79.2	80.0 (+0.8)	81.0 (+1.8)	83.4 (+4.2)
MultiRC	IndE	80.9	81.6	82.5 (+0.9)	81.0 (−0.6)	84.4 (+2.8)
LogicBenchMCQ	AAVE	76.0	72.5	74.0 (+1.5)	76.5 (+4.0)	78.9 (+6.4)
LogicBenchMCQ	IndE	74.8	72.2	64.5 (−7.7)	72.5 (+0.3)	75.3 (+3.1)
FOLIO	AAVE	56.8	56.0	54.5 (−1.5)	52.5 (−3.5)	56.8 (+0.8)
FOLIO	IndE	56.4	57.0	59.5 (+2.5)	61.5 (+4.5)	65.0 (+8.0)

Table 3: Dialect accuracy (%) for GPT-4o under all four prompting conditions. Baseline SAE shown for reference. Δ denotes change from baseline dialect accuracy (pp). Bold denotes improvements of 3.0 pp or more above the baseline dialect accuracy.

Task	Dialect	SAE	Base.	D-Aware (Δ)	Para. (Δ)	F-Shot (Δ)
WSC	AAVE	59.7	52.8	58.5 (+5.7)	58.0 (+5.2)	64.5 (+11.7)
WSC	IndE	58.9	46.0	37.0 (−9.0)	27.5 (−18.5)	56.6 (+10.6)
MultiRC	AAVE	86.0	85.6	86.0 (+0.4)	90.0 (+4.4)	87.9 (+2.3)
MultiRC	IndE	83.5	83.8	83.0 (−0.8)	82.0 (−1.8)	82.9 (−0.9)
LogicBenchMCQ	AAVE	79.3	76.9	76.5 (−0.4)	76.5 (−0.4)	79.0 (+2.1)
LogicBenchMCQ	IndE	75.2	74.3	75.0 (+0.7)	73.0 (−1.3)	84.8 (+10.5)
FOLIO	AAVE	66.8	65.2	60.5 (−4.7)	63.5 (−1.7)	64.8 (−0.4)
FOLIO	IndE	73.8	70.1	70.5 (+0.4)	61.5 (−8.6)	71.5 (+1.4)

Table 4: Dialect accuracy (%) for Gemini 2.5 Pro under all four prompting conditions ($N = 200$). Baseline SAE shown for reference. Δ denotes change from baseline dialect accuracy (pp). Bold denotes improvements of 3.0 pp or more above the baseline dialect accuracy.

Task	Dialect	SAE	Base.	D-Aware (Δ)	Para. (Δ)	F-Shot (Δ)
WSC	AAVE	87.0	80.0	88.0 (+8.0)	89.0 (+9.0)	87.0 (+7.0)
WSC	IndE	89.0	59.0	64.0 (+5.0)	54.0 (−5.0)	63.6 (+4.6)
MultiRC	AAVE	94.0	93.0	95.0 (+2.0)	92.0 (−1.0)	93.9 (+0.9)
MultiRC	IndE	91.0	89.0	90.0 (+1.0)	92.0 (+3.0)	89.9 (+0.9)
LogicBenchMCQ	AAVE	90.0	84.0	89.0 (+5.0)	84.0 (0.0)	85.0 (+1.0)
LogicBenchMCQ	IndE	84.0	82.0	84.0 (+2.0)	78.0 (−4.0)	84.8 (+2.8)
FOLIO	AAVE	80.0	78.0	79.0 (+1.0)	77.0 (−1.0)	74.7 (−3.3)
FOLIO	IndE	85.0	81.0	82.0 (+1.0)	84.0 (+3.0)	82.0 (+1.0)

Dialect-aware instruction framing. Dialect-aware instruction framing reveals a clear dependence on model scale, dialect, and task type. On GPT-4o, the intervention occasionally produces gains, including a substantial improvement for WSC AAVE (+5.7 pp, 83% gap closure) and complete closure on MultiRC AAVE (+0.4 pp, 100%). However, these gains are not consistent across tasks and dialects; for example, performance drops sharply on WSC IndE (−9.0 pp, −69%). While smaller in terms of raw performance degradation, disparity increases were also observed on MultiRC IndE (−0.8 pp, −63%), LogicBenchMCQ AAVE (−0.4 pp, −17%), and FOLIO AAVE (−4.7 pp, −100%).

On GPT-4o-mini, dialect-aware prompting shows mixed results at smaller magnitudes. It produces modest improvements for MultiRC (AAVE: +0.8 pp, 31%; IndE: +0.9 pp, 56%) and LogicBenchMCQ AAVE (+1.5 pp, 43%), but introduces degradations for WSC AAVE (−1.6 pp, −100%), FOLIO AAVE (−1.5 pp, −19%), and especially LogicBenchMCQ IndE (−7.7 pp, −100%). These results suggest that weaker models may struggle to incorporate dialect awareness into their reasoning without degrading accuracy.

Gemini 2.5 Pro shows the strongest overall response to dialect-aware prompting. It fully closes the gap for WSC AAVE (+8.0 pp, 100%) and MultiRC AAVE (+2.0 pp, 100%); it also achieves strong improvements for LogicBenchMCQ AAVE (+5.0 pp, 83%) and LogicBenchMCQ IndE (+2.0 pp, 100%). However, gains remain uneven across dialects, with limited improvement on WSC IndE (+5.0 pp, 17%). Overall, dialect-aware prompting can substantially reduce disparities in favorable settings, but lacks consistency across tasks and dialects.

Dialect normalization. Dialect normalization (paraphrasing into SAE) produces highly inconsistent results and frequently degrades both performance and fairness. On GPT-4o, paraphrasing leads

Table 5: Gap closure (%) toward SAE performance for all models and interventions. Values are capped at $\pm 100\%$. Negative values indicate increased disparity.

Task	Dialect	GPT-4o-mini			GPT-4o			Gemini 2.5 Pro		
		D-Aware	Para.	F-Shot	D-Aware	Para.	F-Shot	D-Aware	Para.	F-Shot
WSC	AAVE	-100	12	100	83	75	100	100	100	100
WSC	IndE	0	-36	100	-69	-100	82	17	-17	15
MultiRC	AAVE	31	69	100	100	100	58	100	-100	90
MultiRC	IndE	56	-17	100	-63	-100	-100	50	100	45
LogicBenchMCQ	AAVE	43	100	100	-17	-17	88	83	0	17
LogicBenchMCQ	IndE	-100	12	100	78	-100	100	100	-100	100
FOLIO	AAVE	-19	-44	10	-100	-100	-25	50	-50	-100
FOLIO	IndE	36	64	100	11	-100	38	25	75	25

to severe drops for WSC IndE (-18.5 pp, -100%) and FOLIO IndE (-8.6 pp, -100%), indicating substantial widening of disparities. Even for smaller drops in raw performance, such as MultiRC IndE (-1.8 pp, -100%), the corresponding gap closure values show a large degradation in fairness.

GPT-4o-mini exhibits similar instability: while paraphrasing improves MultiRC AAVE (+1.8 pp, 69%) and fully closes the gap for LogicBenchMCQ AAVE (+4.0 pp, 100%), it increases disparities for WSC IndE (-1.0 pp, -36%) and FOLIO AAVE (-3.5 pp, -44%).

Gemini 2.5 Pro also shows mixed behavior. While paraphrasing improves WSC AAVE (+9.0 pp, 100%) and MultiRC IndE (+3.0 pp, 100%), it harms WSC IndE (-5.0 pp, -17%) and severely degrades LogicBenchMCQ IndE (-4.0 pp, -100%).

The divergence between Δ and gap closure under paraphrase highlights that even small drops in raw accuracy can correspond to large increases in disparity when baseline gaps are small. These results provide strong evidence that normalization introduces semantic drift that disproportionately harms dialectal inputs, particularly for tasks requiring precise coreference or logical structure.

Few-shot dialect exposure. Few-shot prompting is the most consistently beneficial intervention across models, achieving both large performance gains and strong gap closure. On GPT-4o, it yields substantial improvements for WSC (AAVE: +11.7 pp, 100%; IndE: +10.6 pp, 82%) and LogicBenchMCQ IndE (+10.5 pp, 100%). GPT-4o-mini shows similarly strong gains, fully closing the gap in multiple conditions (e.g., WSC AAVE: +6.9 pp, 100%; LogicBenchMCQ AAVE: +6.4 pp, 100%) and achieving strong improvements for MultiRC and FOLIO IndE.

Gemini 2.5 Pro also benefits from few-shot prompting, achieving complete gap closure for WSC AAVE (+7.0 pp, 100%) and LogicBenchMCQ IndE (+2.8 pp, 100%), as well as strong improvement on MultiRC AAVE (+0.9 pp, 90%). However, gains are not universal, with degradation seen on FOLIO AAVE (-3.3 pp, -100%), indicating limitations under certain conditions including where baseline performance is already high and where dialect gaps are minimal.

Importantly, few-shot prompting rarely produces large negative gap closure compared to other interventions, making it both the most effective and most robust method overall.

6.2 CASE STUDIES: WHEN INTERVENTIONS HARM

While the aggregate results quantify overall trends, they do not reveal the mechanisms underlying the failure modes of each intervention strategy. In this section, we present representative case studies that illustrate how and when each intervention may degrade performance.

Case 1: Dialect-aware instruction on WSC AAVE (GPT-4o-mini): spurious feature interference. For WSC AAVE, an overall performance drop of -1.6 pp and a gap closure of -100% was observed under dialect-aware instruction with GPT-4o-mini.

Figure 1 provides an example of how an explicit dialect label can alter the model’s interpretation in a way that harms reasoning. Under the baseline condition, the model correctly responds with "0", identifying that the pronoun "it" (Span 2) does not refer to "The bag of flour" (Span 1). However,

when prompted with dialect-aware instructions, the model produces an incorrect answer. This failure suggests that the dialect label introduces spurious signals that the model overweights, disrupting its reasoning process rather than helping it adapt.

Baseline Check if Span 2 refers to Span 1 in the paragraph. Paragraph: The sack of potatoes was put on top of the bag of flour, so it gotta be moved first. Span 1: The bag of flour Span 2: it Answer (1 if same, 0 if not): Model's Answer: 0	
Dialect-aware You are given a paragraph written in the African American Vernacular English (AAVE) dialect. Check if Span 2 refers to Span 1 in the paragraph. Paragraph: The sack of potatoes was put on top of the bag of flour, so it gotta be moved first. Span 1: The bag of flour Span 2: it Answer (1 if same, 0 if not): Model's Answer: 1	

Figure 1: **Dialect-aware prompting introduces spurious feature interference in WSC AAVE.** Adding an explicit dialect label causes GPT-4o-mini to misresolve a pronoun that is correctly handled under the baseline condition, resulting in a -1.6 pp performance drop and -100% gap closure.

Case 2: Normalization on FOLIO IndE (GPT-4o): semantic drift through paraphrasing. Normalization via paraphrasing can subtly alter the meaning of the input, leading to incorrect predictions. For FOLIO IndE, this corresponds to an overall performance drop of -8.6 pp and a gap closure of -100% with GPT-4o.

As shown in Figure 2, the paraphrased version modifies a key logical structure: the statement "not both a Ferrari and made in Maranello" is paraphrased to "neither a Ferrari nor manufactured in Maranello." While superficially similar, this change alters the logical relationship between predicates, leading to an incorrect inference ("False") compared to the baseline condition's correct inference ("Uncertain"). This illustrates how paraphrasing can introduce semantic drift particularly in tasks that depend on precise logical structure, where even minor changes in wording can invalidate the reasoning required to solve the task.

Case 3: Few-shot prompting on FOLIO AAVE (Gemini-2.5-Pro): overcommitment bias from demonstrations. While few-shot prompting is generally beneficial, its failure cases reveal a consistent pattern: when degraded performance is observed, it is usually caused by the model abandoning "Uncertain" conclusions in favor of decisive "True" or "False" predictions. For FOLIO AAVE, this corresponds to an overall performance drop of -3.3 pp and a gap closure of -100% with Gemini 2.5 Pro.

In Figure 3, the model correctly identifies that the conclusion is "Uncertain" under the baseline condition. However, after exposure to few-shot examples, the model produces an incorrect "False" judgement.

Notably, this pattern holds across all observed few-shot failure cases under both Gemini 2.5 Pro and GPT-4o: whenever performance degrades, the error involves the model abandoning an "Uncertain" conclusion in favor of a definitive "True" or "False" conclusion. This suggests that few-shot prompting can introduce a systematic bias toward decisive answers in more powerful models.

This behavior reflects a form of overgeneralization from demonstrations: the model learns to mimic the structure and confidence of the examples, even when the underlying reasoning does not transfer.

Baseline Determine whether the following conclusion is true, false, or uncertain based on the premises. Answer with only True, False, or Uncertain. Premises: Hermès bags are not being made in Italy. All Birkin bags are Hermès bags only. All Ferraris are being made in Italy. All cars which are carrying a Ferrari V12 engine are Ferraris. All cars which are made in Maranello are carrying a Ferrari V12 engine. A Lamborghini SUV is not both a Ferrari and made in Maranello. A Kelly bag is either a Hermès bag or it is a car that is carrying a Ferrari V12 engine. Conclusion: A Lamborghini SUV is made in Italy. Answer: Model's Answer: Uncertain
Paraphrased Determine whether the following conclusion is true, false, or uncertain based on the premises. Answer with only True, False, or Uncertain. Premises: Hermès bags are not manufactured in Italy. Every Birkin bag is exclusively a Hermès bag. All Ferraris are manufactured in Italy. Every car equipped with a Ferrari V12 engine is a Ferrari. Every car manufactured in Maranello is equipped with a Ferrari V12 engine. A Lamborghini SUV is neither a Ferrari nor manufactured in Maranello. A Kelly bag is either a Hermès bag or a car equipped with a Ferrari V12 engine. Conclusion: A Lamborghini SUV is made in Italy. Answer: Model's Answer: False

Figure 2: **Paraphrase-based normalization introduces semantic drift in FOLIO IndE.** A subtle change from "not both A and B" to "neither A nor B" alters the logical structure of the premise, leading GPT-4o to change its answer from "Uncertain" to "False". This results in a -8.6 pp performance drop and -100% gap closure, demonstrating how normalization can increase disparity by distorting task-critical semantics.

As a result, few-shot prompting can degrade both accuracy and fairness by systematically reducing the model’s ability to represent uncertainty.

This failure mode highlights an important limitation of few-shot prompting: while it improves performance in many settings, it can also distort the model’s calibration, particularly in tasks requiring careful distinction between decisive and uncertain conclusions.

Summary of failure modes. These failure cases reveal how interventions, meant to mitigate existing sources of error in interpreting dialectal inputs, can also introduce new ones. Dialect-aware prompting may inject irrelevant signals, normalization may distort meaning, and few-shot prompting may encourage overfitting to examples. These findings underscore the importance of evaluating interventions not only by their average performance gains, but also by their worst-case behaviors and their respective impacts on existing disparities.

Baseline

Determine whether the following conclusion is true, false, or uncertain based on the premises. Answer with only True, False, or Uncertain.

Premises:

Some monitors they got in the library be made by AOC.

All the monitors they got in the library cost less than 800 dollars.

All the monitors that cost less than 800 dollars got a resolution lower than 1080p.

If a monitor got a resolution lower than 1080p, then it ain't supportin' the type-c port.

A-2017 support the type-c port.

Conclusion: A-2017 is produced by AOC.

Answer:

Model's Answer: Uncertain

Few-Shot

Determine whether the following conclusion is true, false, or uncertain based on the premises. Answer with only True, False, or Uncertain.

Here are a few examples:

Premises: All folks who drink coffee on the regular be hooked on caffeine. People either drink coffee a lot, or they ain't tryna be addicted to caffeine, or it's both. Ain't nobody who don't wanna be addicted to caffeine unaware that caffeine a drug. Rina either a student who don't know caffeine a drug, or she ain't a student and she know caffeine a drug. Rina either a student who hooked on caffeine, or she ain't a student and ain't hooked on caffeine.

Conclusion: Rina doesn't want to be addicted to caffeine or is unaware that caffeine is a drug.

Answer: True

[Additional examples omitted for brevity]

Now answer the following:

Premises:

Some monitors they got in the library be made by AOC.

All the monitors they got in the library cost less than 800 dollars.

All the monitors that cost less than 800 dollars got a resolution lower than 1080p.

If a monitor got a resolution lower than 1080p, then it ain't supportin' the type-c port.

A-2017 support the type-c port.

Conclusion: A-2017 is produced by AOC.

Answer:

Model's Answer: False

Figure 3: **Few-shot prompting induces overcommitment under uncertainty in FOLIO AAVE.** While the baseline model correctly identifies the conclusion as "Uncertain", few-shot prompting leads the model to predict "False". This reflects a consistent failure mode in which demonstrations bias the model toward definitive answers, reducing its ability to represent uncertainty and increasing disparity.

7 CHALLENGES AND SOLUTIONS

In this section, we highlight the challenges we faced in our evaluation and their associated limitations, as well as corresponding solutions.

Small or inconsistent dialect gaps on recent models. For several task–dialect–model combinations, the observed performance gap relative to SAE is small, near zero, or even negative. This may reflect improvements in recent model generations (GPT-4o, Gemini 2.5 Pro) compared to the models originally tested in EnDive. Demonstrating the ability of interventions to reduce an observed gap becomes more difficult when the observed gap is already small.

In these cases, even when interventions produce measurable changes in accuracy (Δ , pp), the corresponding gap closure values can be unstable or misleading. Small absolute changes can yield large positive or negative gap closure percentages when the baseline gap is small.

Solution: We jointly analyze both Δ (pp) and gap closure (%) to distinguish raw performance changes from meaningful reductions in disparity. We focus our qualitative analysis on tasks with larger baseline gaps (e.g., WSC, LogicBenchMCQ), where fairness improvements are more interpretable.

Paraphrase drift in normalization. Dialect normalization relies on a generative model to rewrite dialectal inputs into SAE, introducing the risk that task-critical linguistic structure is inadvertently altered. This is particularly problematic for WSC, where pronoun antecedent relationships can shift under paraphrase, and for formal logic tasks, where the precise wording of premises and quantifiers determines the correct inference.

Solution: We interpret normalization results in light of each task’s sensitivity to precise wording and use the case studies to illustrate concretely where and why paraphrase drift harms performance. These findings motivate future work on constrained normalization that explicitly preserves task-critical structure.

Inconsistent effectiveness of interventions. Across models, tasks, and dialects, all three interventions exhibit substantial variability in their impact on both performance and fairness. While few-shot prompting is the most consistently beneficial, it still fails in certain settings (e.g., FOLIO AAVE), producing negative Δ and gap closure due to overcommitment to decisive predictions. Dialect-aware prompting shows even greater variability, yielding strong improvements in some cases (e.g., WSC AAVE for GPT-4o and Gemini) while increasing disparities in others (e.g., WSC IndE and LogicBenchMCQ IndE for GPT-4o-mini). Paraphrasing is the least reliable intervention, frequently introducing semantic drift that leads to large negative gap closure values, particularly in grammatically sensitive settings.

This inconsistency is reflected in the divergence between Δ and gap closure. Interventions that appear to produce small performance changes can still substantially increase or decrease disparities, depending on the underlying gap. As a result, no single intervention uniformly improves dialect accuracy across all settings.

Solution: Rather than treating interventions as universally beneficial, we evaluate them as context-dependent strategies whose effectiveness varies by model capacity, task structure, and dialect. By jointly reporting Δ and gap closure, we identify where interventions meaningfully reduce disparities and where they introduce new failure modes. These findings suggest that future work should focus on adaptive or task-specific intervention strategies rather than universal approaches.

Computational cost and evaluation limits. Evaluating large-scale models such as Gemini 2.5 Pro introduces practical constraints due to higher computational cost and latency compared to smaller models. As a result, we restrict evaluation for Gemini 2.5 Pro to a subset of $N = 200$ examples per task–dialect condition, rather than the full set of EnDive tasks used for GPT-4o and GPT-4o-mini. While this enables tractable experimentation, it introduces additional variance in both Δ and gap closure estimates, particularly for tasks with smaller baseline gaps where individual prediction changes can have a significant impact on overall results.

Solution: We interpret Gemini 2.5 Pro results with appropriate caution, focusing on consistent qualitative patterns across tasks rather than small numerical differences. Where possible, we corroborate trends with results from GPT-4o and GPT-4o-mini, which are evaluated on larger samples.

8 CONCLUSION

We investigated whether three inference-time prompt interventions—dialect-aware instruction framing, dialect normalization, and few-shot dialect exposure—can reduce the performance gaps that LLMs exhibit on dialectal English reasoning tasks. Evaluating GPT-4o-mini, GPT-4o, and Gemini 2.5 Pro on WSC, MultiRC, LogicBenchMCQ, and FOLIO in AAVE and IndE, we find that: (1) dialect performance gaps are largest on WSC IndE (up to 30 pp for Gemini 2.5 Pro) and consistent but smaller on logic tasks (1–6 pp); (2) few-shot prompting is the most robust intervention, achieving full gap closure in multiple conditions and rarely producing large negative effects; (3) normalization is highly task-sensitive, helping on tasks where surface dialect variation can be removed without altering meaning (e.g., MultiRC, WSC AAVE for Gemini 2.5 Pro) but severely harming tasks requiring precise wording (e.g., WSC IndE for GPT-4o: −18.5 pp); and (4) dialect-aware instruction framing is strongly model-dependent, benefiting Gemini 2.5 Pro most consistently while producing mixed or negative results for GPT-4o-mini.

Taken together, our results paint a cautionary picture for practitioners. A developer seeking to deploy an LLM equitably across dialect groups cannot rely on a single prompt intervention: the intervention that helps on one task-dialect combination may significantly degrade another. Few-shot prompting comes closest to a safe default, but even it provides minimal benefit when baseline gaps are already small or negative. Normalization, despite its intuitive appeal as a pre-processing step, introduces semantic drift that invalidates task-critical structure in pronoun disambiguation and formal logic problems, making it a risky choice without task-specific validation.

These findings also carry a broader fairness implication. The speakers most affected by dialect performance gaps, approximately 30 million AAVE speakers and 129 million IndE speakers, are often in contexts where retraining or fine-tuning is not a realistic option for the systems they interact with. If prompt-level interventions are insufficient, this places the burden of equitable access squarely on the model training pipeline itself, motivating dialect-inclusive pretraining data and dialect-robust evaluation as first-class concerns in LLM development.

Promising future directions include combining interventions (e.g., dialect-aware instruction paired with few-shot examples), conducting per-feature analysis on specific dialect structures such as zero copula, habitual *be*, and existential *it*, extending evaluation to other English dialects such as Chicano English, Jamaican English, and Colloquial Singaporean English, and exploring lightweight dialect-robust decoding strategies as a step beyond prompting.

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